

Jon Arrizabalaga

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🔗 [jonarriza96.github.io](https://github.com/jonarriza96)

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Education

PhD	Technical University of Munich , Robotics & Control • Thesis , Defense	2020 – 2025
MSc	KTH Royal Institute of Technology , Mechatronics Engineering • Thesis	2018 – 2020
BSc	University of Navarre , Mechanical Engineering	2014 – 2018

Academic Experience

Massachusetts Institute of Technology , Postdoctoral Researcher	2025 – present
Carnegie Mellon University , Visiting PhD / Postdoctoral Researcher	2024 – 2026
Charles University , Visiting PhD	2023
Shanghai Jiao Tong University , Visiting Student	2017 – 2018

Industry Experience

Bosch Research , MSc. Thesis - Robotics Researcher • Paper , Video	2020
Porsche AG , Research Intern	2019 – 2019

Honors & Awards

Best Robocup Paper Award (finalist) 5 nominations out of ~4000 submissions at the conference IEEE/RJS IROS • Paper , Talk	2024
Best PhD Lecturer Award Recognizes the PhD lecturer who has received the highest evaluations from BSc and MSc students	2022
Outstanding End-of-Degree Award (finalist) Honors BSc. students who have excelled in both their academic and extracurricular endeavors	2018

Invited Talks

Northeast Systems and Control Workshop , Columbia University	2025
Talking Robotics , Virtual Seminar • Talk	2025
GRASP Laboratory , Vijay Kumar Lab, University of Pennsylvania	2024
German Aerospace Center, DLR , Institute of Flight Systems	2023
Robotics and Perception Group , University of Zurich	2022

Teaching

Engineering Mechanics I , Lecturer, Technical University of Munich <i>Level: BSc, Students: 200 ~ 450</i>	2021 – 2024
Mechanics for Aerospace , Lecturer, Technical University of Munich (Asia) <i>Level: MSc, Students: 20 ~ 25</i>	2021 – 2023
Introduction to ROS (Robot Operating System) , Lecturer, Technical University of Munich <i>Level: MSc, Students: 50</i>	2021 – 2022
Dynamics Motion and Control , Teaching Assistant, KTH Royal Institute of Technology <i>Level: MSc, Students: 50</i>	2019 – 2020
Robust Mechatronics , Teaching Assistant, KTH Royal Institute of Technology <i>Level: MSc, Students: 50</i>	2019 – 2020

Publications

DQ-NMPC: Dual-Quaternion NMPC for Quadrotor Flight • Paper , Video , Code , Website	2025
L.F. Recalde, D. Agrawal, <i>J. Arrizabalaga</i> , G. Li, RA-L	
Convex Maneuver Planning for Spacecraft Collision Avoidance • Paper , Code	2025
F. Vega, <i>J. Arrizabalaga</i> , R. Watson, Z. Manchester, IEEE iSpaRo	
The Trajectory Bundle Method: Unifying Sequential-Convex Programming and Sampling-Based Trajectory Optimization • Paper	2025
K. Tracy, J. Zhang, <i>J. Arrizabalaga</i> , S.Schaal, Y. Tassa, T. Erez, Z. Manchester, Under Review	
Conflict-Based Search as a Protocol: A Multi-Agent Motion Planning Protocol for Heterogeneous Agents, Solvers, and Independent Tasks • Paper , Website	2025
R. Veerapaneni,, A. Tang, H. He, S. Zhao, V. Shah, Y. Cen, Z. Ji, G. Olin, <i>J. Arrizabalaga</i> , Y. Shaoul, M. Likhachev Likhachev, Under Review	
A Universal Formulation for Path-Parametric Planning and Control • Paper , Code , Website , Talk	2025
<i>J. Arrizabalaga</i> , Z. Sir, Z. Manchester, M. Ryll, Under Review	
PHODCOS: Pythagorean Hodograph-based Differentiable Coordinate System • Paper , Code , Talk	2025
<i>J. Arrizabalaga</i> , F. Vega, Z. Sir, Z. Manchester, M. Ryll, IEEE Aero	
Differentiable Collision-Free Parametric Corridors • <i>Best Paper Finalist</i> • Paper , Code , Video , Talk	2024
<i>J. Arrizabalaga</i> , Z. Manchester, M. Ryll, IEEE IROS	
Geometric Slesh-Free Tracking for Robotic Manipulators • Paper , Code , Video , Talk	2024
<i>J. Arrizabalaga</i> , L. Pries, R. Laha, R. Li, S. Haddadin, M. Ryll, IEEE ICRA	
Learning for CasADi: Data-Driven Models in Numerical Optimization • Paper , Code , Talk	2023
T. Salzmann, <i>J. Arrizabalaga</i> , J. Andersson, M. Pavone, M. Ryll, L4DC	
Pose-Following with Dual Quaternions • Paper , Code , Talk	2023
<i>J. Arrizabalaga</i> , M. Ryll, IEEE CDC	
SCTOMP: Spatially Constrained Time-Optimal Motion Planning • Paper , Video	2023
<i>J. Arrizabalaga</i> , M. Ryll, IEEE IROS	
Spatial Motion Planning with Pythagorean Hodograph curves • Paper , Video , Talk	2022
<i>J. Arrizabalaga</i> , M. Ryll, IEEE CDC	
Towards Time-Optimal Tunnel-Following for Quadrotors • Paper , Video	2022
<i>J. Arrizabalaga</i> , M. Ryll, IEEE ICRA	
Real-Time Neural-MPC: Deep Learning Model Predictive Control for Quadrotors and Agile Robotic Platforms • Paper , Code	2022
T. Salzmann, E. Kaufmann, <i>J. Arrizabalaga</i> , M. Pavone, D. Scaramuzza, M. Ryll, RA-L	
A caster-wheel-aware MPC-based motion planner for mobile robotics • Paper , Thesis , Video	2021
<i>J. Arrizabalaga</i> , M. Ryll, IEEE ICAR	

Students Mentored

Viraj Shah , Research Project, <i>now at</i> Carnegie Mellon University	2025
Eden Cen , Research Project, <i>now at</i> Carnegie Mellon University	2025
Akshay Jaitly , Research Project, <i>now at</i> Worcester Polytechnic Institute	2025
Luis Fernando Recalde , Research Project, <i>now at</i> Worcester Polytechnic Institute	2024
Giuseppe Soldati , Semester Thesis, <i>now at</i> German Aerospace Center (DLR)	2023
Harun Tongay Tamtürk , Semester Thesis & MSc. Thesis, <i>now at</i> Airbus Defence and Space	2023
Bruno Sorban , Research Project, <i>now at</i> Boston Consulting Group	2023
Tommaso Faraci , Research Project, <i>now at</i> Universita di Trento	2022

Extracurricular Projects

Ascent - Rocket Hopper , Guidance and Control Lead • Website	2022 – 2024
The Juggling Robot Project , Team Lead • Video	2019 – 2020
Formula Student Electric , Vehicle Dynamics Engineer	2016 – 2018

Review Activities

- IEEE Transactions on Robotics (T-RO)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Systems, Man, and Cybernetics: Systems (T-SMC)
- Robotics Science and Systems (RSS)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Conference on Decision and Control (CDC)
- IEEE International Conference on Humanoid Robots (Humanoids)
- IEEE International Conference on Space Robotics (iSpaRo)